

Integrated Circuit ECE326

Lec. 15

Layout

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- ☐ **Layout Methodologies**
- ☐ **Circuit Elements**
- ☐ **Resistor**
- ☐ **Capacitor**
- ☐ **Inductor**
- ☐ **MOSFET**
- ☐ **CMOS Inverter**
- ☐ **Series MOSFETs**
- ☐ **Parallel MOSFETs**
- ☐ **NAND Gate**
- ☐ **NOR Gate**

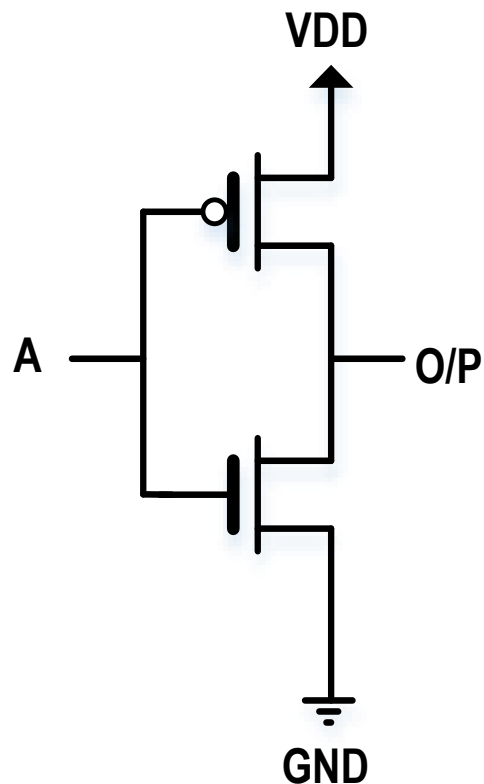


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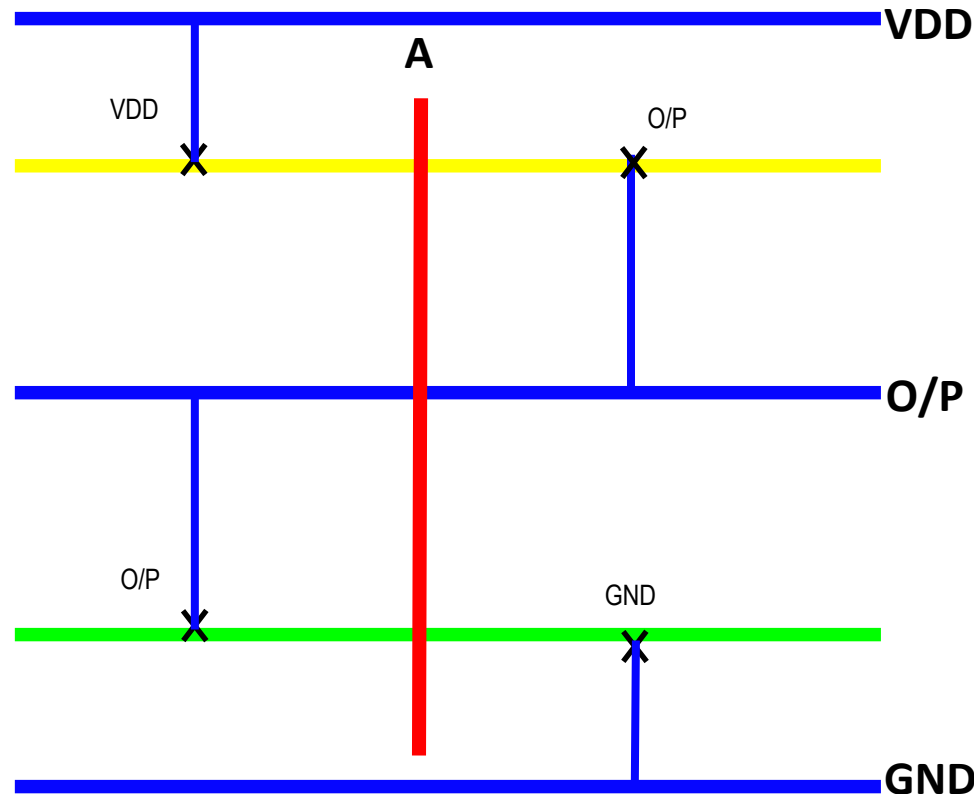
Layout Methodologies



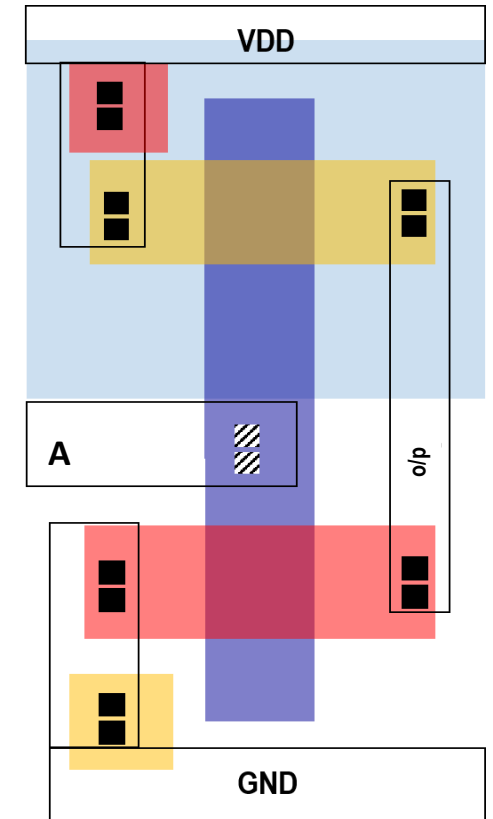
- ❑ Layout can be very time consuming to fit gates/elements together nicely
 - ❑ Stick diagram
 - ❑ Standard cell design methodology



Schematic Circuit



Stick Diagram



Standard Cell design

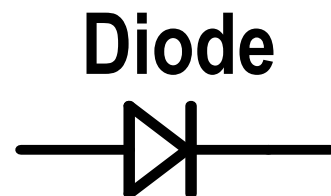
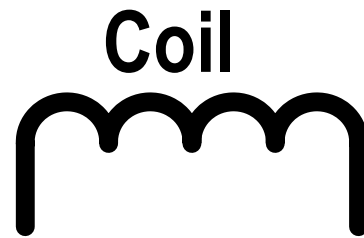
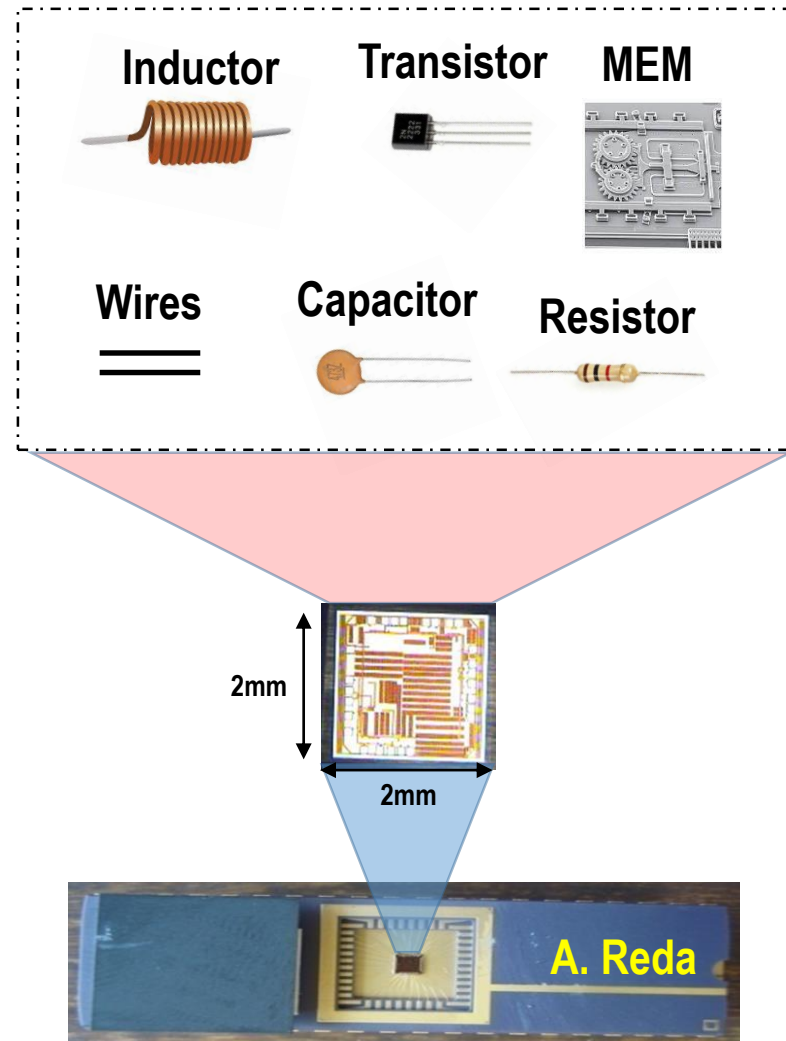


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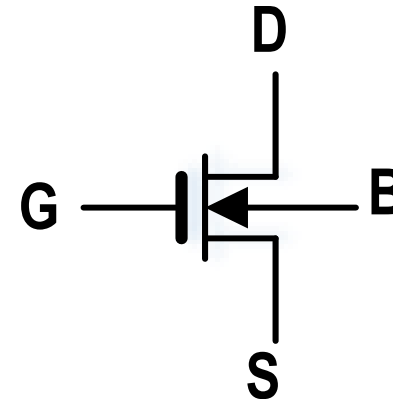
Circuit Elements



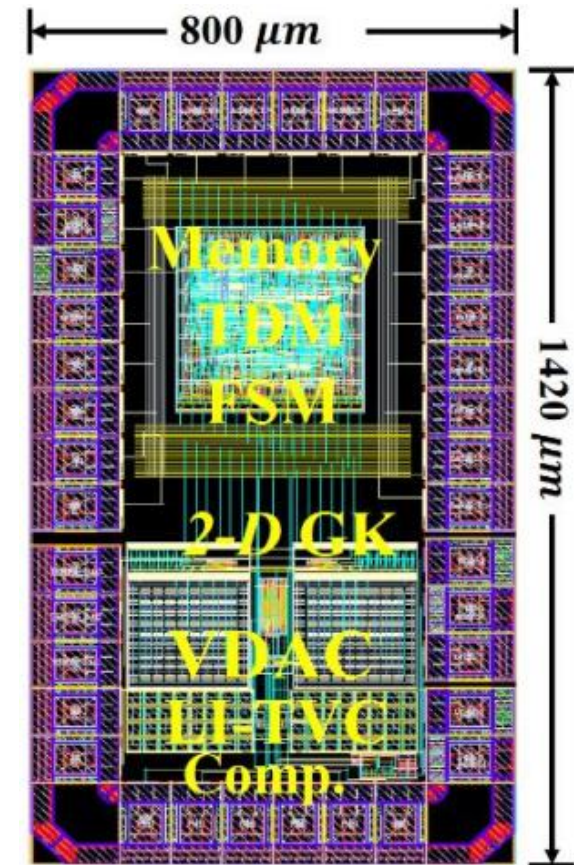
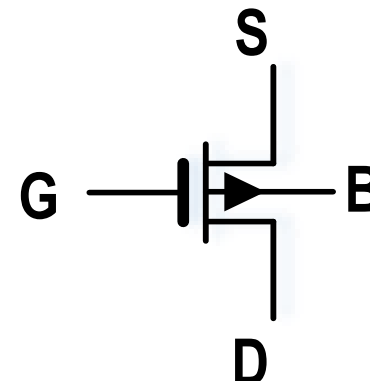
❑ Main electronic elements on-chip are resistor, capacitor, diode, and MOSFET.



N-MOSFET



P-MOSFET



Layout using CAD tool



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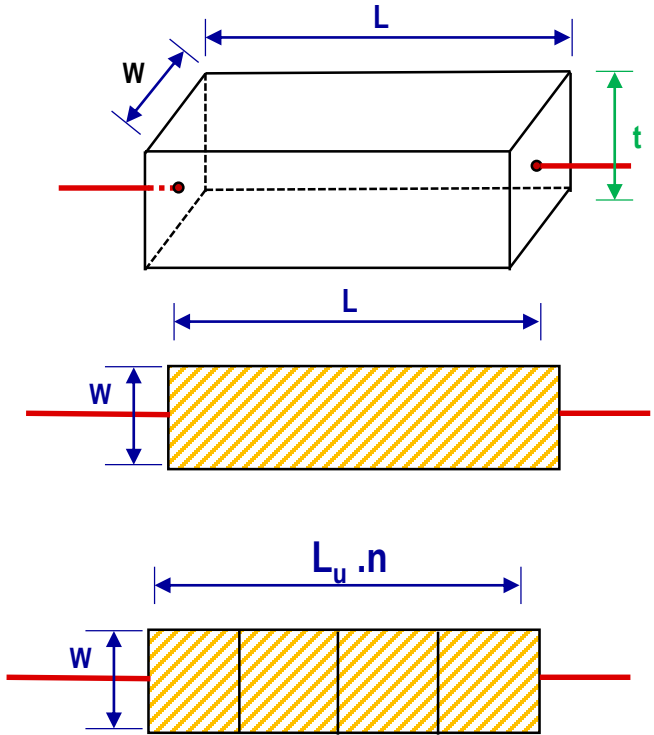
Resistor (1/7)

$R = \rho \cdot \frac{L}{A} = \rho \cdot \frac{L}{t \cdot W} = \rho_{\square} \cdot \frac{L}{W}$ **→** $R = \rho_{\square} \cdot \frac{L}{W}$ & $R = \rho_{\square} \cdot \frac{L_u}{W} \cdot n$

$\rho_{\square}$: depends on material and its characteristics (Poly metal, n-type, p-type,...)

Type of materials		
Item	Poly	Metal
Sheet Resistance	HIGH↑	LOW↓
Consumed silicon area (W.L)	LOW↓	HIGH↑
sensitivity to temp.	HIGH↑	LOW↓
Precision	LOW↓	HIGH↑
Use	HIGH resistor	LOW resistor

Resistance	
Material	Sheet resistance (Ω/□)
N-well	1000
P-well	35
Metal	0.003
Poly	25



- R** : Resistance
- ρ** : Resistivity
- ρ_□** : Sheet resistance (Ω/□)
- L** :Length of resistor
- A** : cross section
- W** : width of resistor
- t** : thickness of resistor

Typical Process →

0.18 μm Process FAB

Device	Name	Available with module	RS [$\Omega/\square$]	Temp. Coeff. [$10^{-3}/\text{K}$]	Max VTB [V]
N+ Poly	rn p1, rnp1_3*	LPMOS	320	-1.38	45
P+ Poly	r pp1, rpp1_3*	LPMOS	280	-0.11	45
P+ Poly silicided	rpp1s, rpp1s_3*	LPMOS	7.5	2.92	45

Device	Name	Available with module	RS [$\Omega/\square$]	Temp. Coeff. [$10^{-3}/\text{K}$]	Max VTB [V]
1.8V N+ diffusion	rdn rdn_esd*	LPMOS HVMOS	65	1.42	1.98
1.8V P+ diffusion	rdp rdp_io*	LPMOS HVMOS	130	1.3	1.98
1.8V N-well	rnw, rnw_scr*	LPMOS	970	2.90	5.5

Device	Name	Available with module	RS [$\Omega/\square$]	Thickness [μm]	Max J/W [mA/ μm]	Temp. Coeff. [$10^{-3}/\text{K}$]	Max VTB [V]
Metal 1	rm 1	LPMOS	0.077	0.555	0.5*	3.4	45
Metal 2	rm2	LPMOS	0.074	0.565	0.5*	3.4	45

Resistor (3/7)



- Ex.1 (a) Find the estimated minimum consumed silicon area to fabricate a resistor of $1\text{k}\Omega$ using $0.13\mu\text{m}$ n-well process, 1LM. (b) Sketch the construction of the resistor on the substrate, (c) Sketch the layout of this resistor.

Given from Process/Technology

Resistance	
Material	Sheet resistance ($\Omega/\square$)
N-well	10
P-well	20
Metal	1
Poly	100

Ans.

Resistor (4/7)



(a) for minimum area $R = \rho_{\square} \uparrow \cdot \frac{L}{W} \downarrow$

Poly material is selected because $\rho_{\square} = 100 \rightarrow$ highest $\rho_{\square}$

$$R = \rho_{\square} \cdot \frac{L}{W}$$

$$1000 = 100 \cdot \frac{L}{W}$$

Assume $W = 0.13 \mu\text{m}$

$L = 1.3 \mu\text{m}$

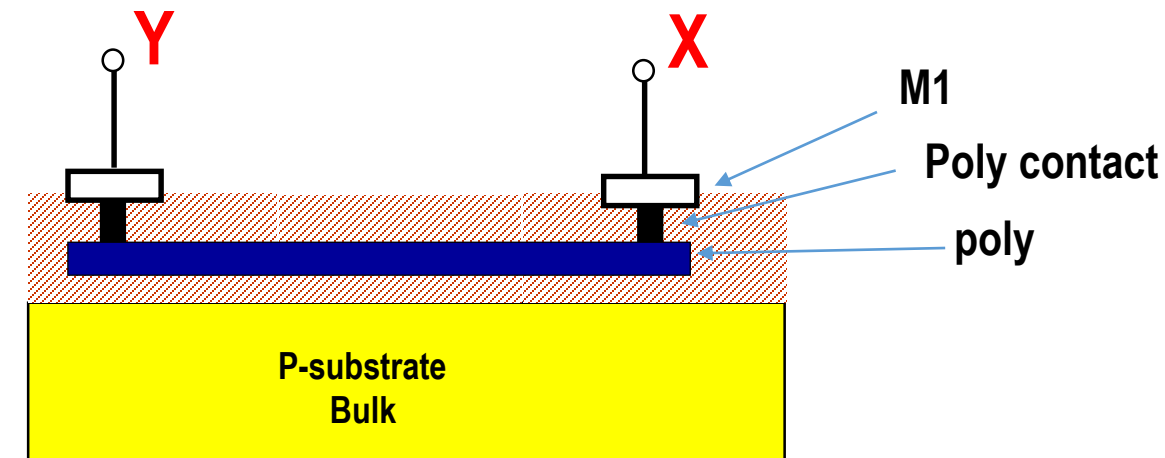
Estimated area = $W \cdot L = 0.13 \cdot 1.3 = \mu\text{m}^2$

Resistor (5/7)



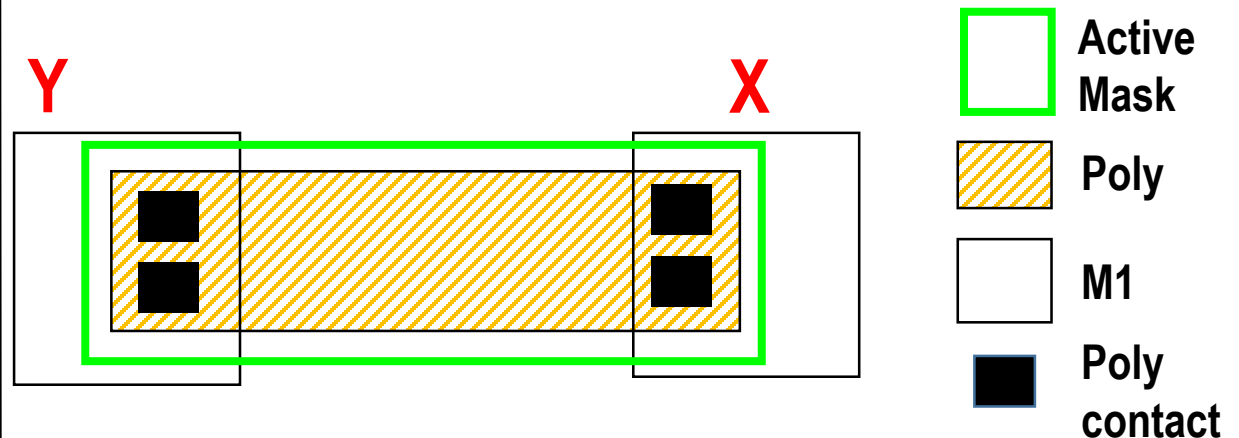
(b) Construction

- ❑ Since the given is n-well process, it means that the p-substrate is provided.
- ❑ Since the type of resistor is Poly (You have determined in (a)), poly resistor will be **on** the substrate.
- ❑ Since the provided process is 1LM, it means that only Metal 1 can be used.



(C) Layout

- ❑ No need to add mask for p-Substrate and even oxidation process.
- ❑ More than one contact is preferred to reduce Ohmic contacts.



Resistor (6/7)



- Ex.2) Calculate the resistance of an n-well that is 10 wide and 100 long. Assuming that the n-well's sheet resistance is typically 2KΩ/square; however, it can vary with process shifts from 1.6 KΩ/square to 2.4 KΩ/square.

Ans.

The typical resistance is $R = \rho_{\square} \cdot \frac{L}{W} = 2000 \cdot \frac{100}{10} = 20 \text{ K}\Omega$

The Maximum resistance is $R = \rho_{\square} \cdot \frac{L}{W} = 2400 \cdot \frac{100}{10} = 24 \text{ K}\Omega$

The Minimum resistance is $R = \rho_{\square} \cdot \frac{L}{W} = 1600 \cdot \frac{100}{10} = 16 \text{ K}\Omega$

Resistor (7/7)



- Ex.3 , (a) Compute the sheet resistance of a 0.22 μm thick Cu wire in a 65 nm process. Where $\rho_{\text{Cu}}=2.2 \mu \Omega \cdot \text{cm}$
- Find the total resistance if the wire is 0.125 μm wide and 1 mm long..

Ans.

Sheet resistance is $\rho_{\square} = \frac{\rho_{\text{Cu}}}{\Delta} = \frac{2.2 * 10^{-8}}{0.22 * 10^{-6}} = 0.1 \Omega / \square$

Total resistance is $R = \rho_{\square} \cdot \frac{L}{W}$

$$R = 0.1 * \frac{1 * 10^{-3}}{0.125 * 10^{-6}} = 800 \Omega$$

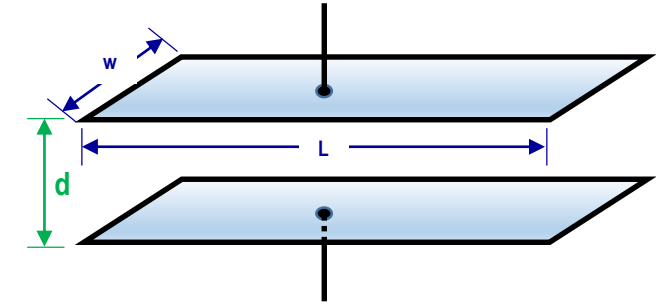


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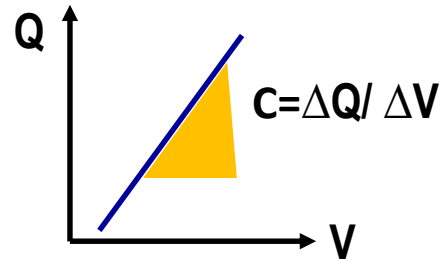
Capacitor (1/6)



❑ Any two conduction materials which are insulated by insulator can be considered as capacitor



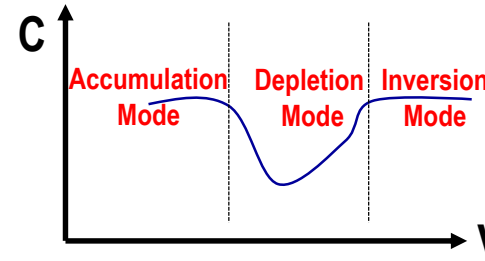
Linear capacitor ☺



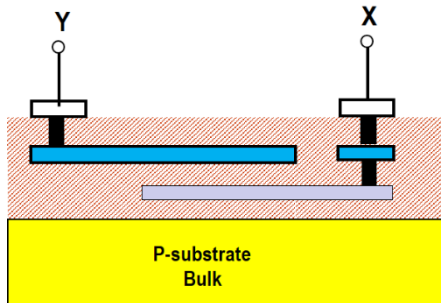
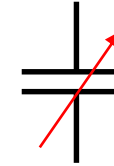
Used for Constant capacitor



Non-linear capacitor



Used for Variable capacitor



Metal-Insulator-Metal (MIM)

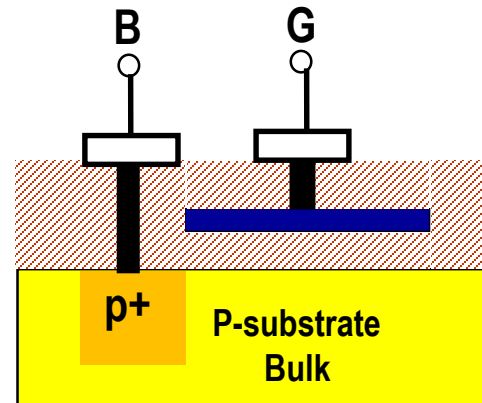
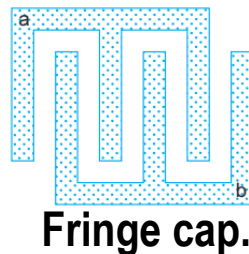
Poly-Insulator -Poly (PIP)

Parallel Plate Capacitor (PPC)

$$C = \epsilon_o \cdot \epsilon_r \cdot \frac{A}{d} = C_o \cdot A$$

$$C = C_o \cdot W \cdot L$$

C_o : Capacitance per unit area



poly-to-semiconductor (MOS capacitor)

$$C \propto V_g$$

Capacitor (2/6)



Typical Process →

0.18 μm Process **X FAB**

XH018 VARACTORS

Device	Name	Available with module	Tuning range [%]	Capacitance @-V [$\text{fF}/\mu\text{m}^2$]	Capacitance @+V [$\text{fF}/\mu\text{m}^2$]	Max VBpsub [V]
1.8V N-type varactor	mosvc	LPMOS	70	2.6	8.1	5.5
3.3V N-type varactor	mosvc3	LPMOS	70	1.7	5	5.5

XH018 FRINGE CAPACITORS

Device	Name	Available with module	Cell Cap [fF]	BV [V]	Max. VTB [V]
Poly1/M1/M2 fringe	csf2p	LPMOS	25.2	> 15	45

XH018 MIM CAPACITOR

Device	Name	Available with module	Area Cap [$\text{fF}/\mu\text{m}^2$]	V Coeff. [1/V]	BV [V]	max. VTB [V]
Single MIM, M2/M3	cmm3	MIM23	1.00	-15	> 20	45
Single MIM, M3/M4	cmm4	MIM34	1.00	-15	> 20	45

Capacitor (3/6)



- Ex.1 (a) Find the estimated minimum consumed silicon area to fabricate a capacitor of 10 fF using 0.13 μm n-well process, 3LM. (b) Sketch the construction of the capacitor on the substrate, (c) Sketch the layout of this capacitor.

Given from Process/Technology

Capacitor	
Material	C_o (fF/ μm^2)
Gate to channel	0.17
Poly to substrate	0.03
Metal1 to Metal2	0.5
Poly to metal 1	0.4

Ans.

Capacitor (4/6)



(a) for minimum area $C=C_o \uparrow .(W.L) \downarrow$

M1 to M2 is selected because $C_o=0.5 \text{ fF}/\mu\text{m}^2 \rightarrow$ highest C_o

$$C=C_o.(W.L)$$

$$10 \text{ fF}=0.5 \text{ fF}/\mu\text{m}^2 (W.L)$$

$$W.L=20 \mu\text{m}^2$$

Assume $w=5 \mu\text{m}$

$L=4 \mu\text{m}$

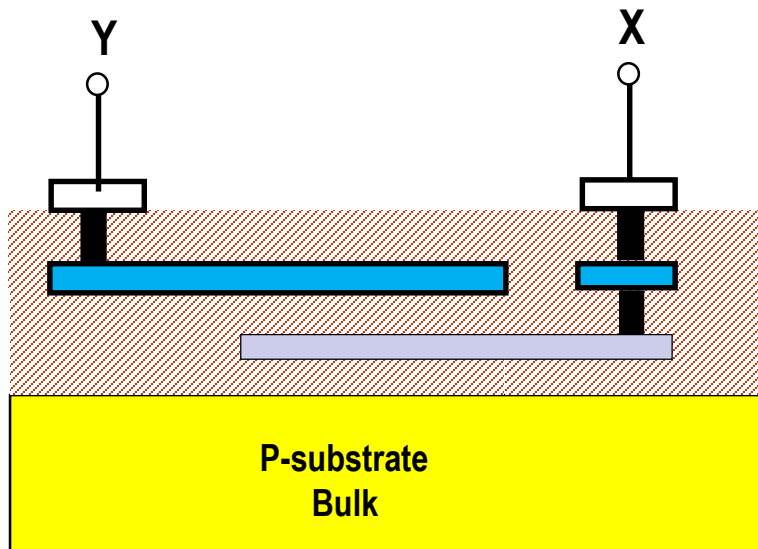
Estimated area= $W.L=4*5=20 \mu\text{m}^2$

Capacitor (5/6)



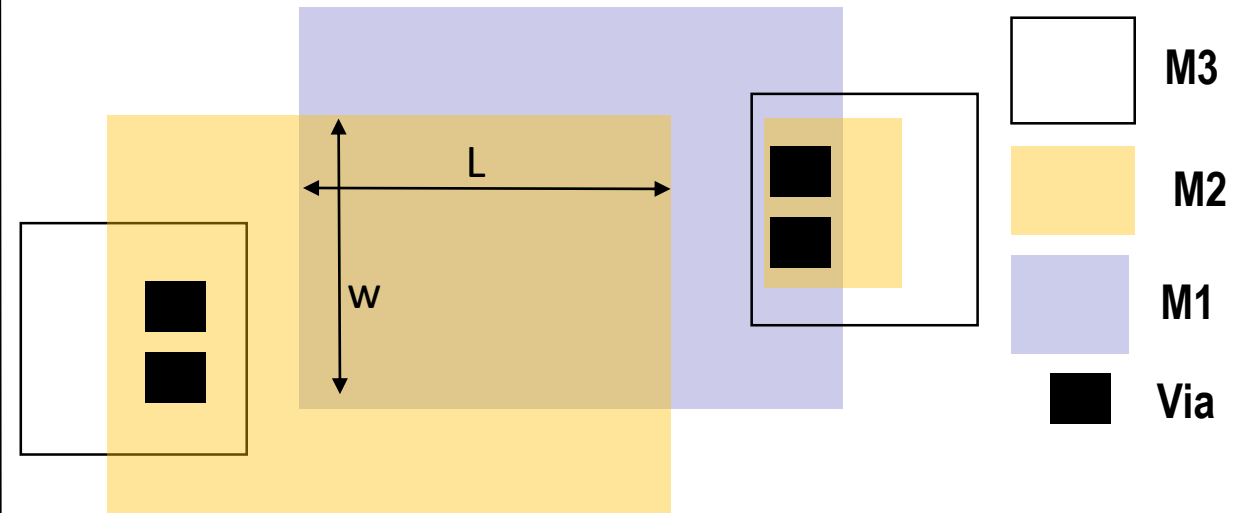
(b) Construction

- ❑ Since the given is n-well process, it mean that the p-substrate is provides.
- ❑ Since the type of capacitor is M1 to M2 (You have determined in (a)), Both M1 and M2 will be **on** the substrate.



(C) Layout

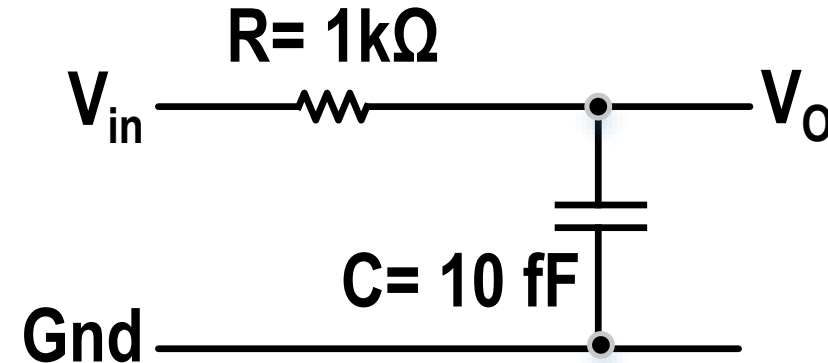
- ❑ No need to add mask for p-Substrate and even oxidation process.
- ❑ More than one contact is preferred to reduce Ohmic contacts.



Assignment (1/3)



- a) Find the estimated minimum consumed silicon area to fabricate the shown circuit using $0.13\mu\text{m}$ n-well process, 3LM. (b) Sketch the construction of the circuit on the substrate, (c) Sketch the layout of this circuit.



Given from Process/Technology

Capacitor	
Material	C_o (fF/ μm^2)
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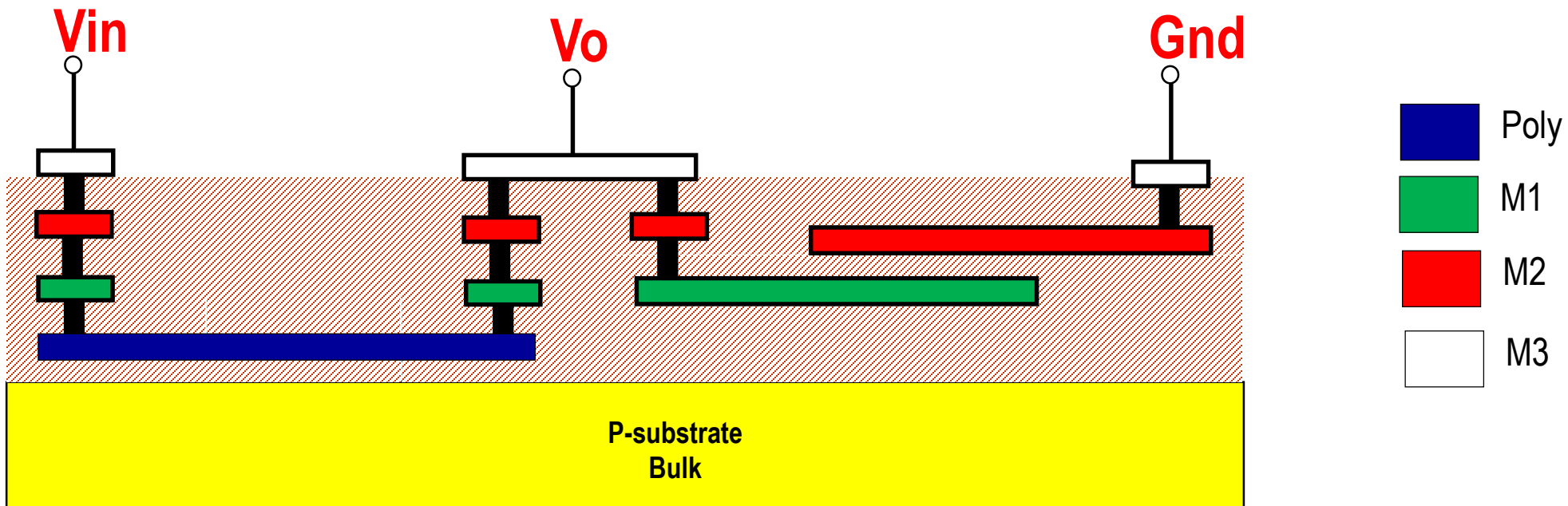
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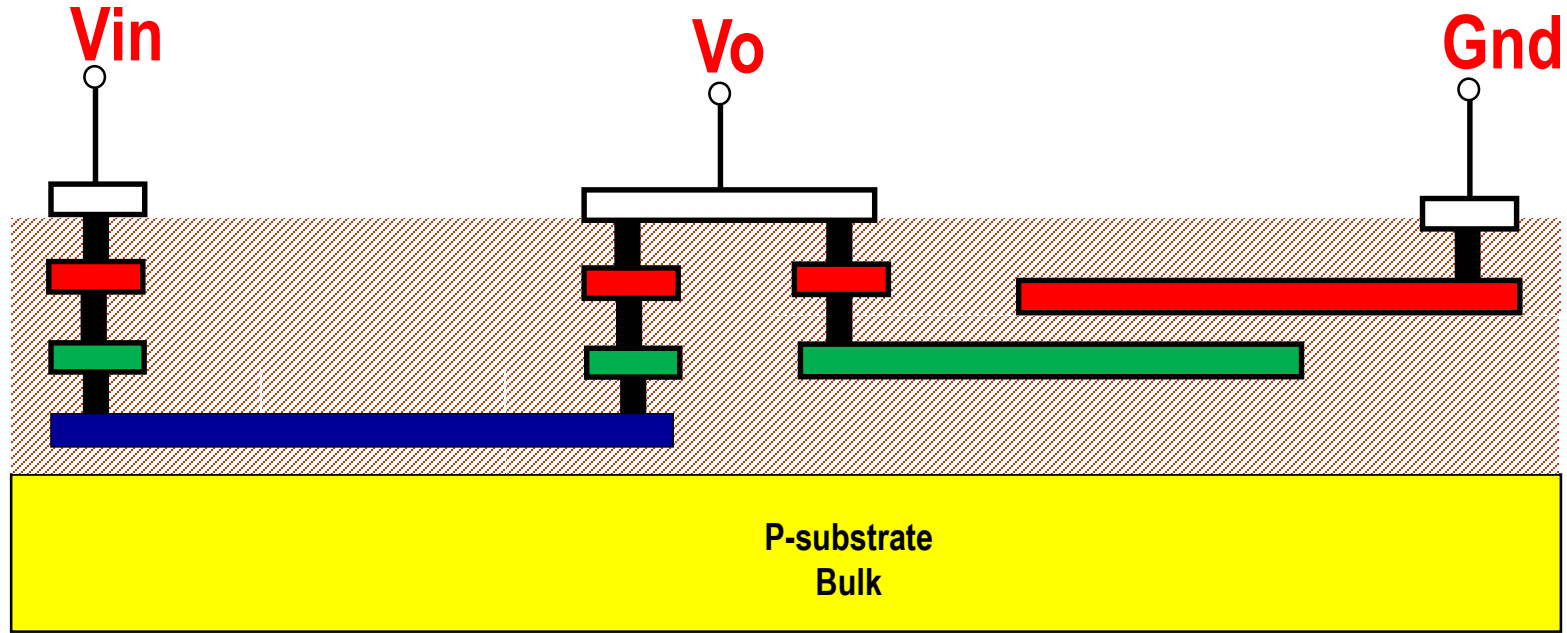
Assignment (2/3)



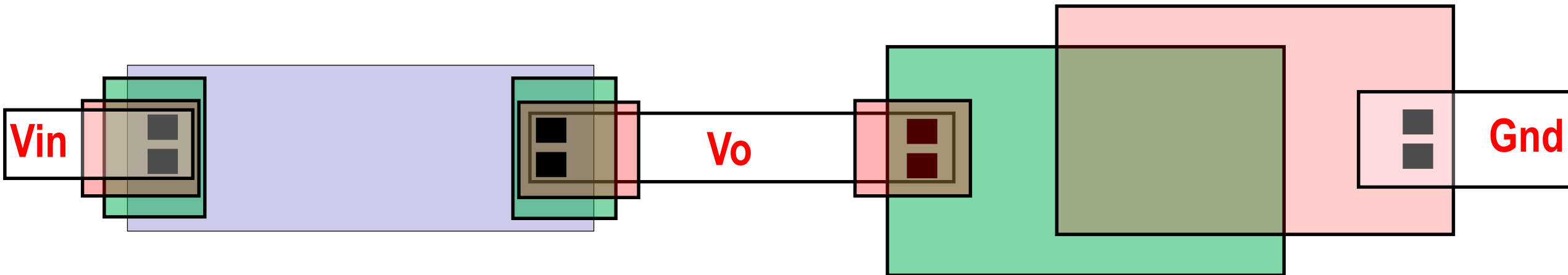
- a) Find the estimated minimum consumed silicon area to fabricate the shown circuit using $0.13\mu\text{m}$ n-well process, 3LM. (b) Sketch the construction of the circuit on the substrate, (c) Sketch the layout of this circuit.



Assignment (3/3)



- Poly
- M1
- M2
- M3
- Contact



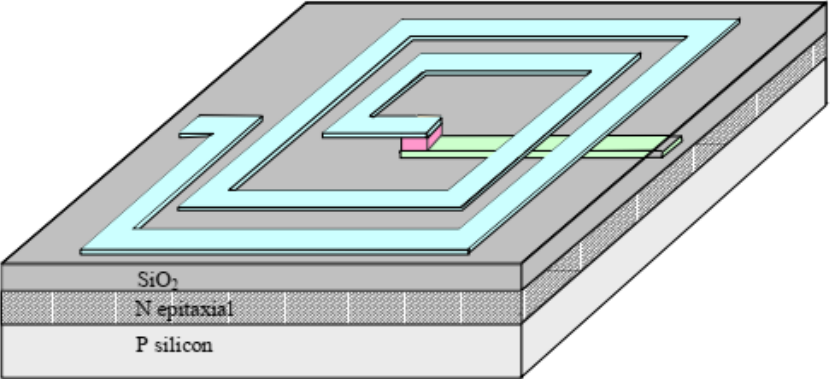
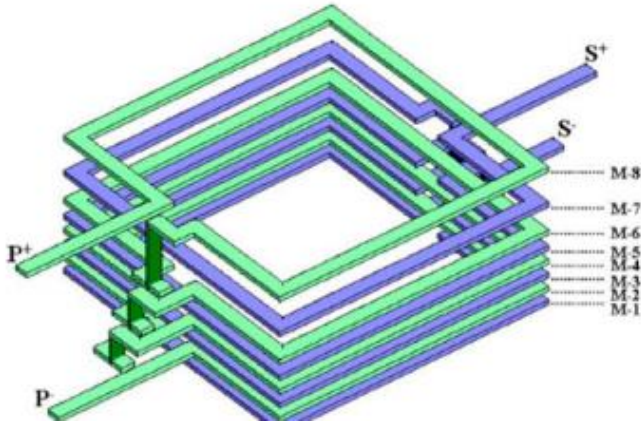


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Inductor



- ❑ The desire to integrate inductors on chips has increased radically with the upsurge in interest in RF circuits
- ❑ The most common monolithic inductor is the spiral inductor, which is a spiral of upper-level metal

Spiral	Coil
❑ Uses the top layer for planer coil	❑ Uses vertical turns with uniform radius
❑ Quality factor (Q) is in range of 5 to 10 	❑ Use small silicon area ❑ Provides large inductance (L) 

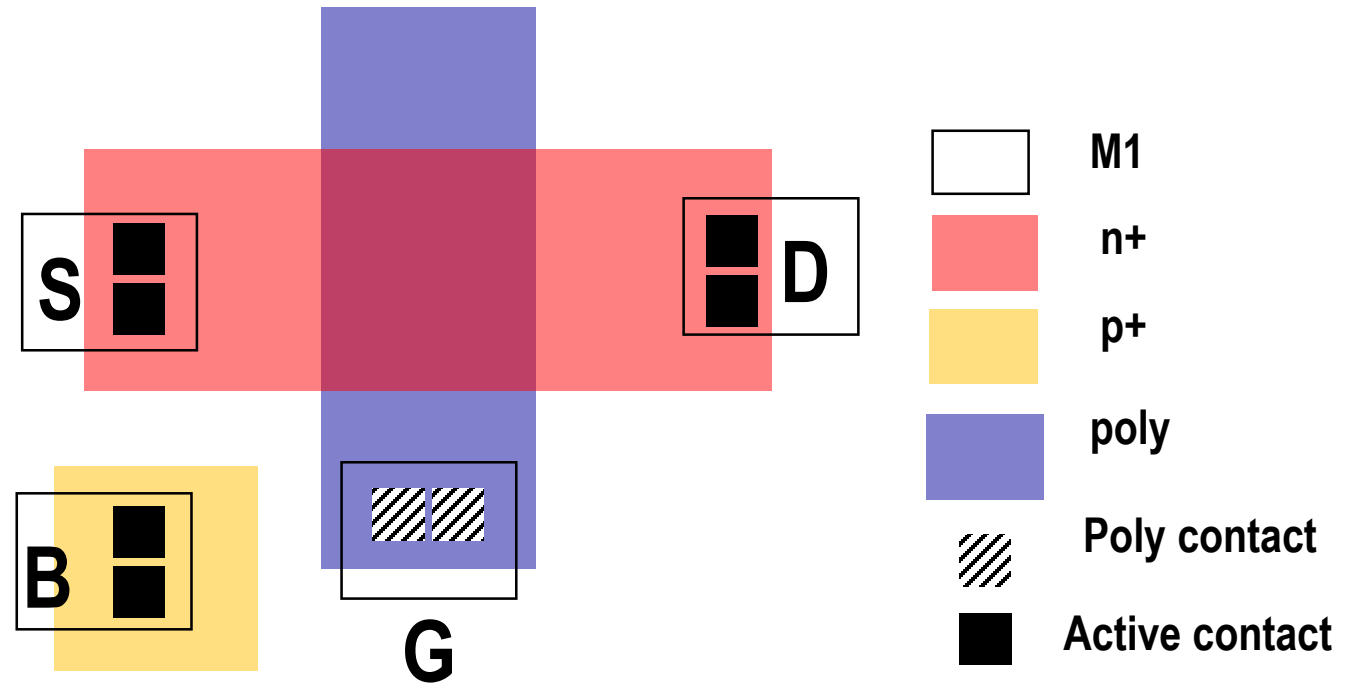
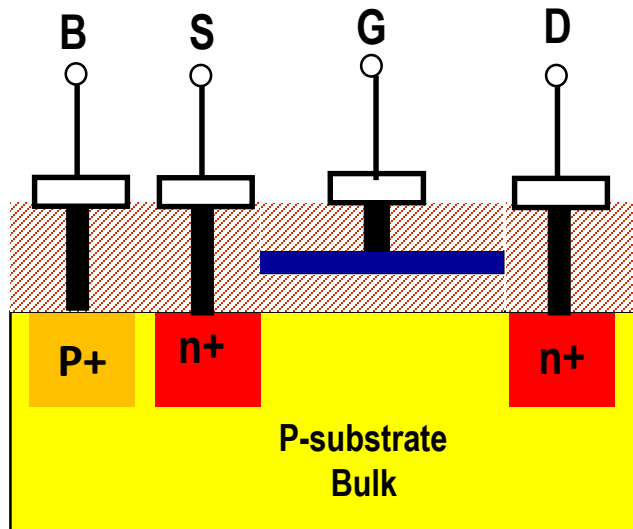
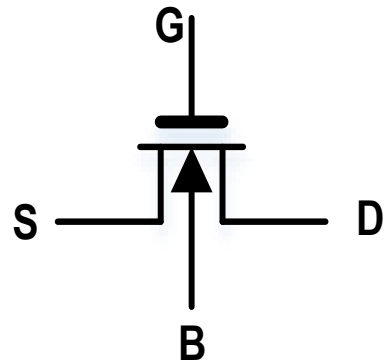


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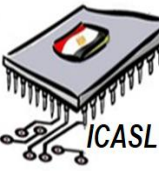
MOSFET (1/2)



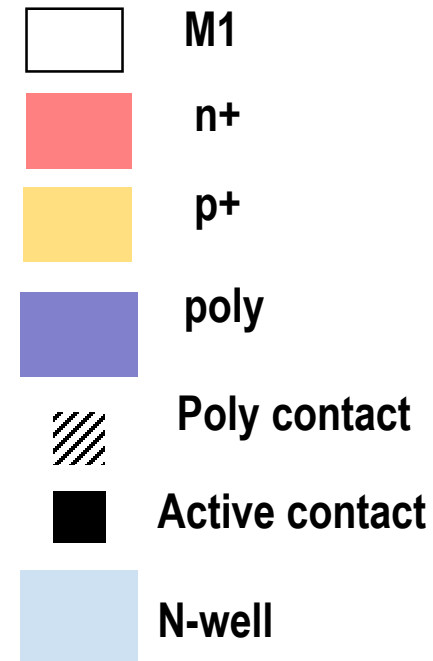
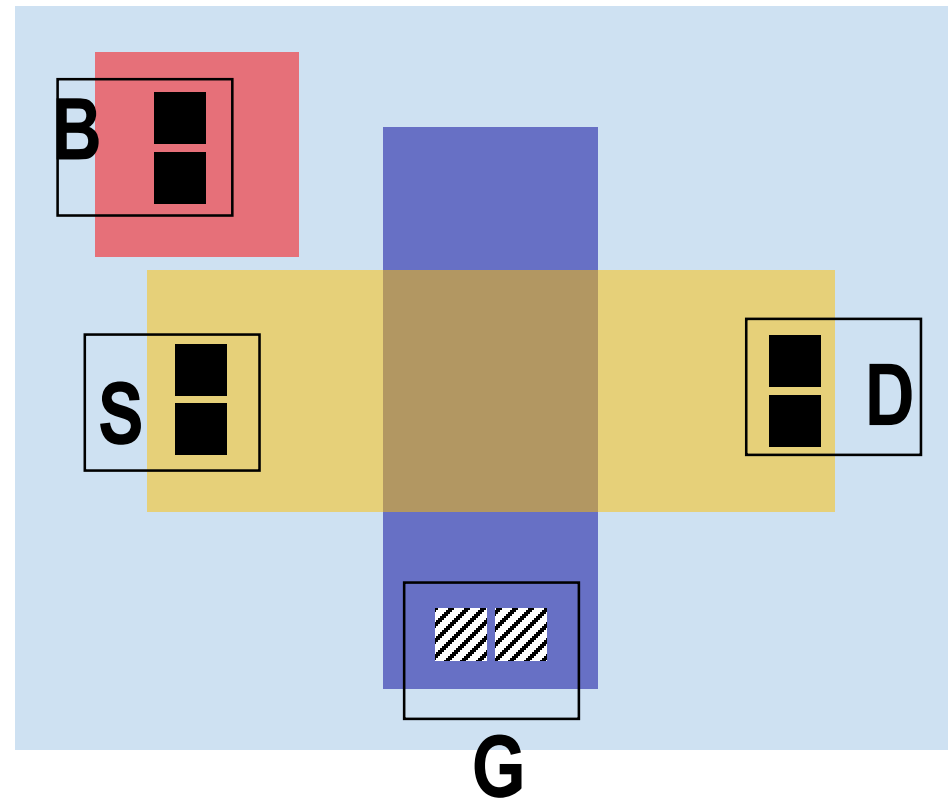
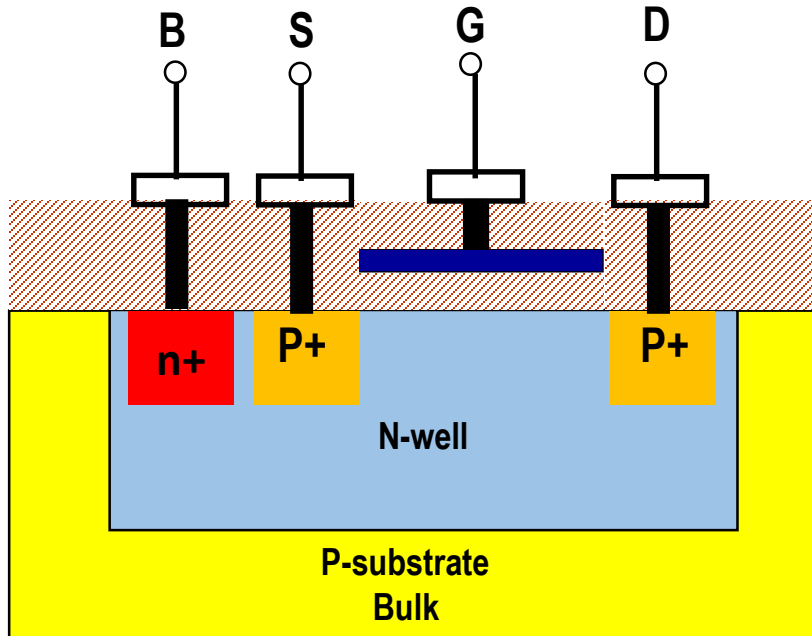
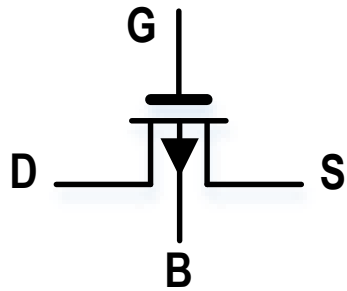
□ n-MOSFET needs p- substrate



MOSFET (2/2)



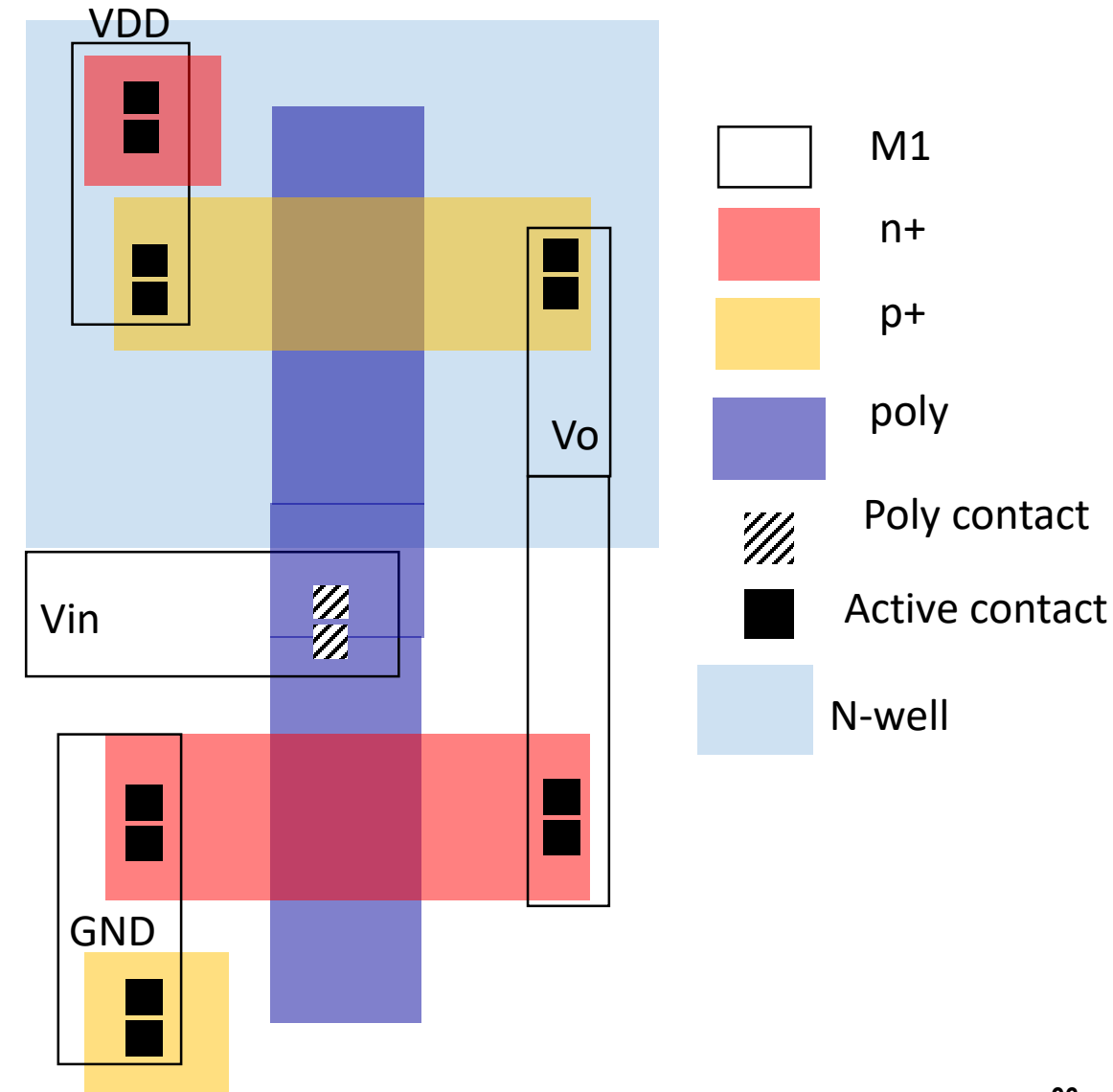
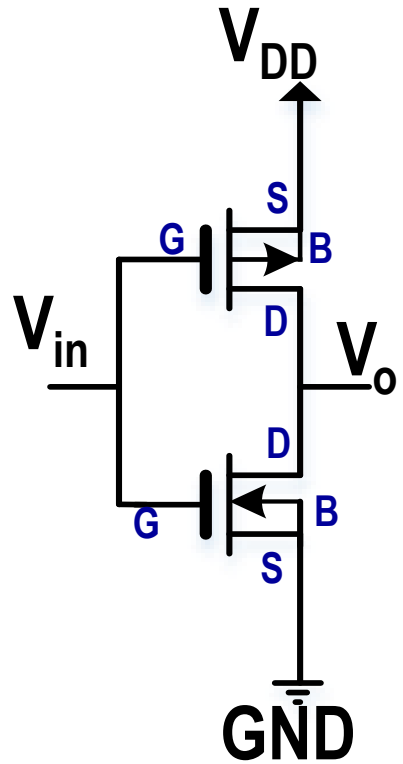
❑ P-MOSFET needs n-well if a p- substrate is provided





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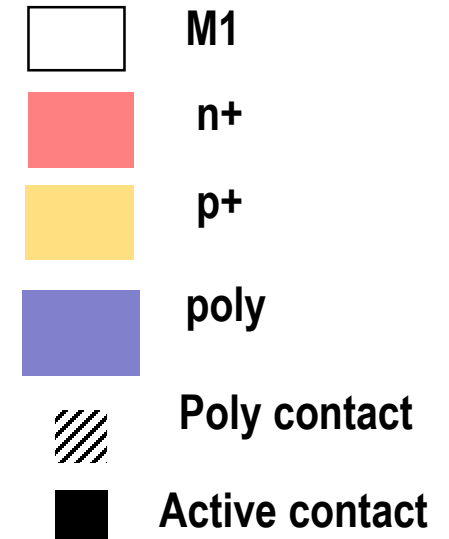
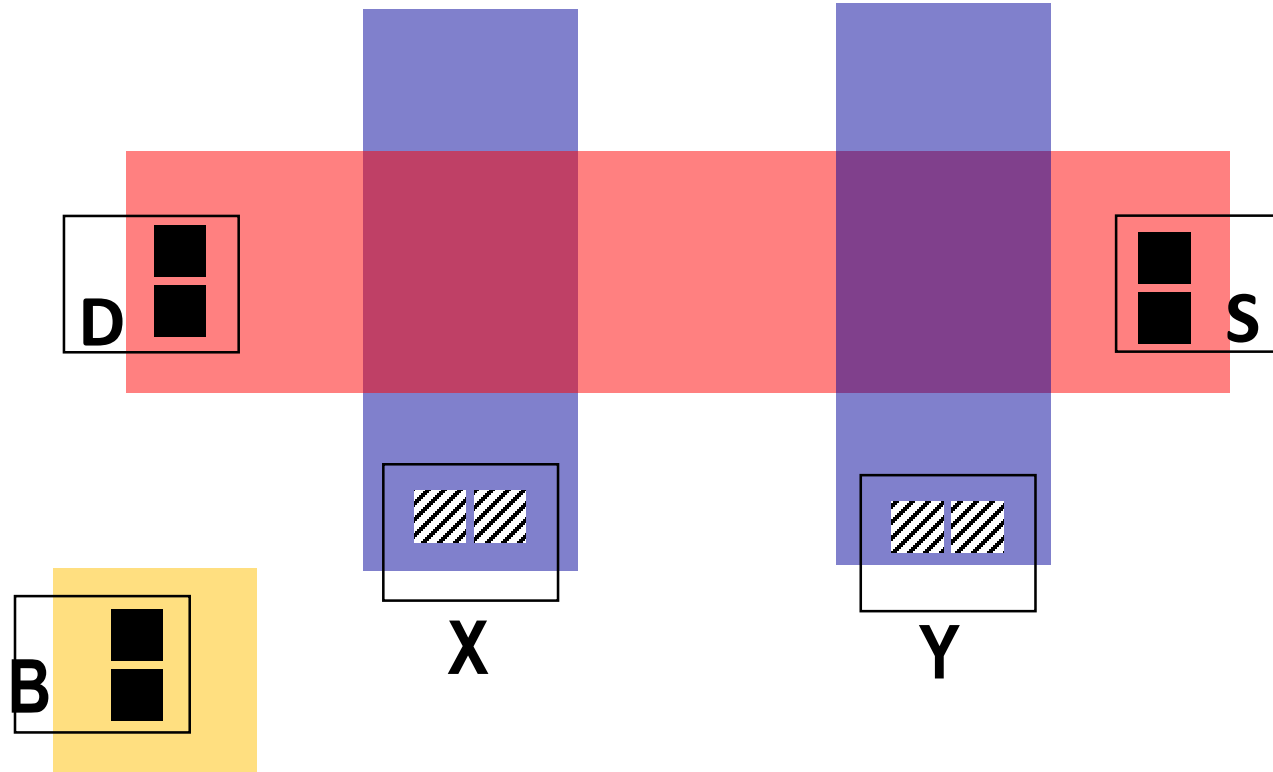
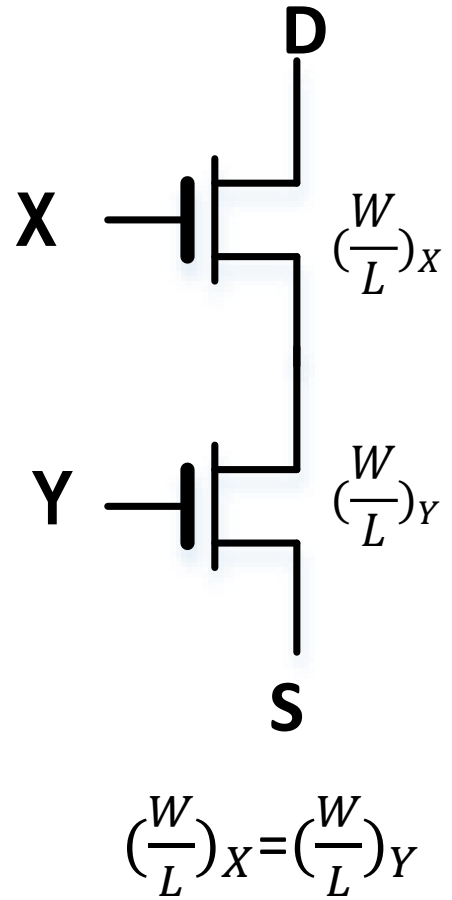
CMOS Inverter



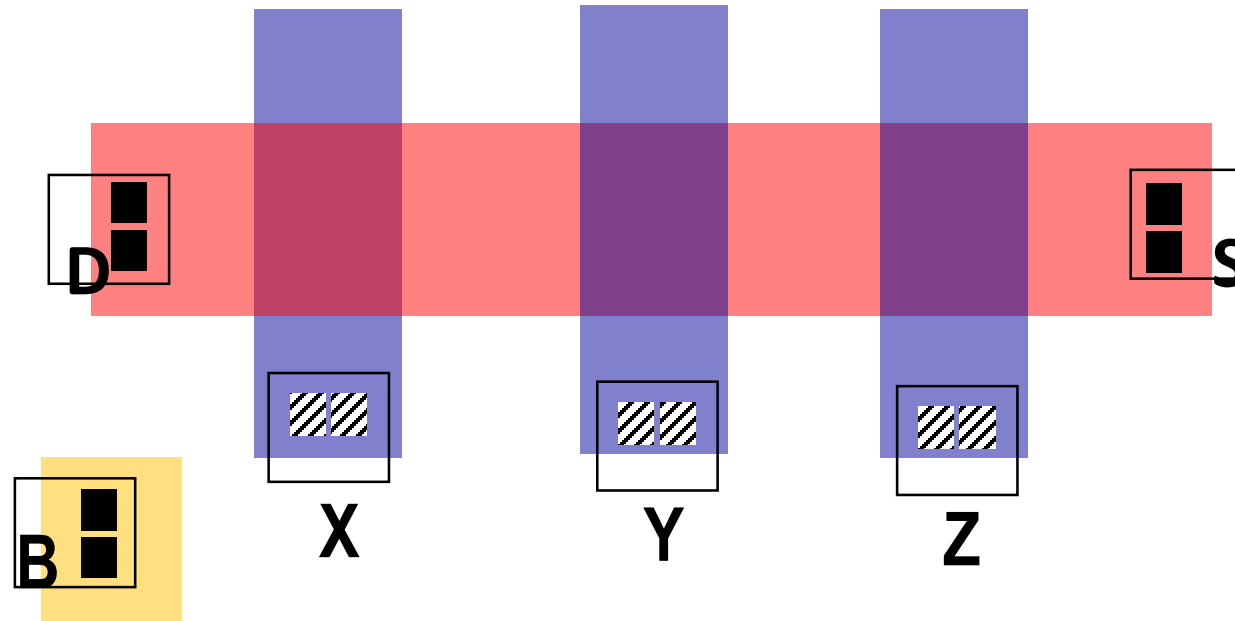
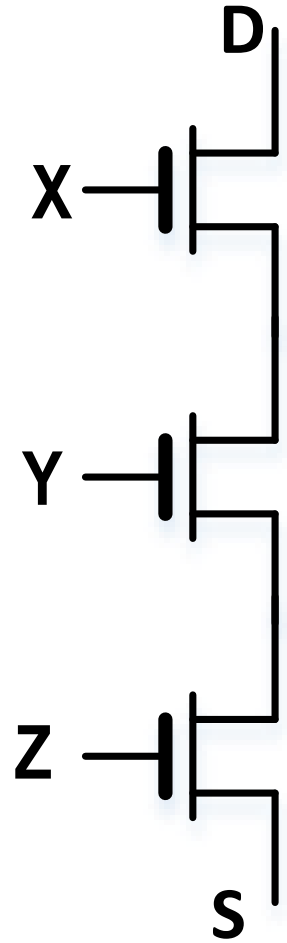


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Series MOSFETs (1/2)



Series MOSFETs (2/2)



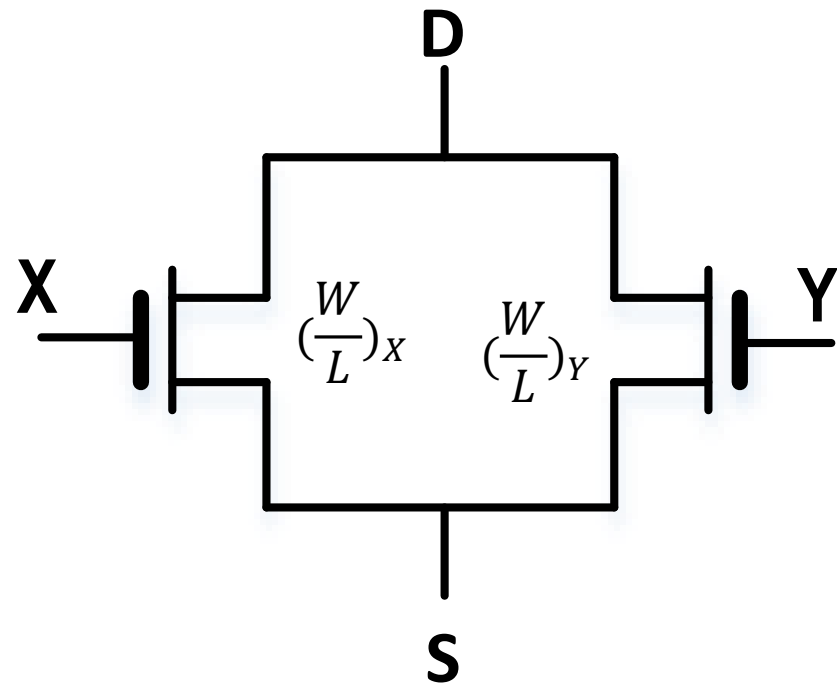
-  M1
-  n+
-  p+
-  poly
-  Poly contact
-  Active contact

$$\left(\frac{W}{L}\right)_X = \left(\frac{W}{L}\right)_Y = \left(\frac{W}{L}\right)_Z$$

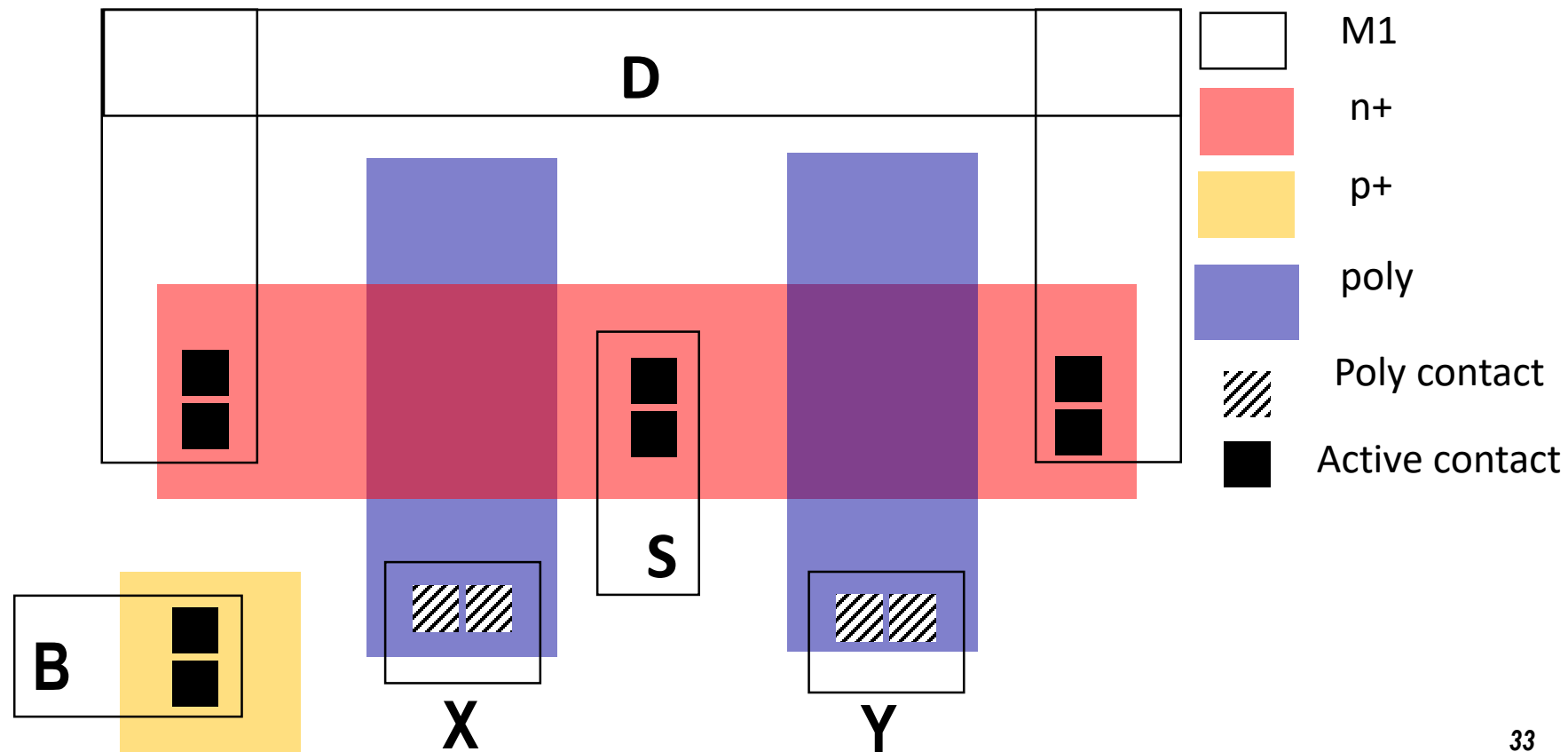


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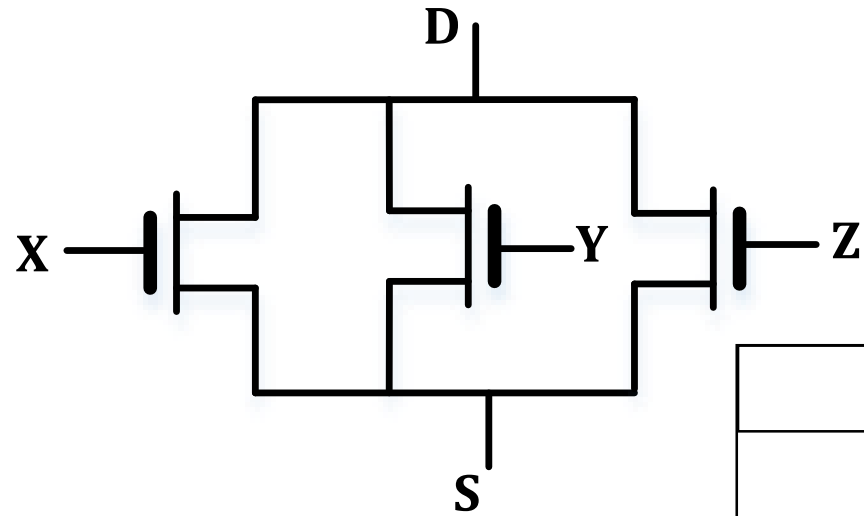
Parallel MOSFETs (1/2)



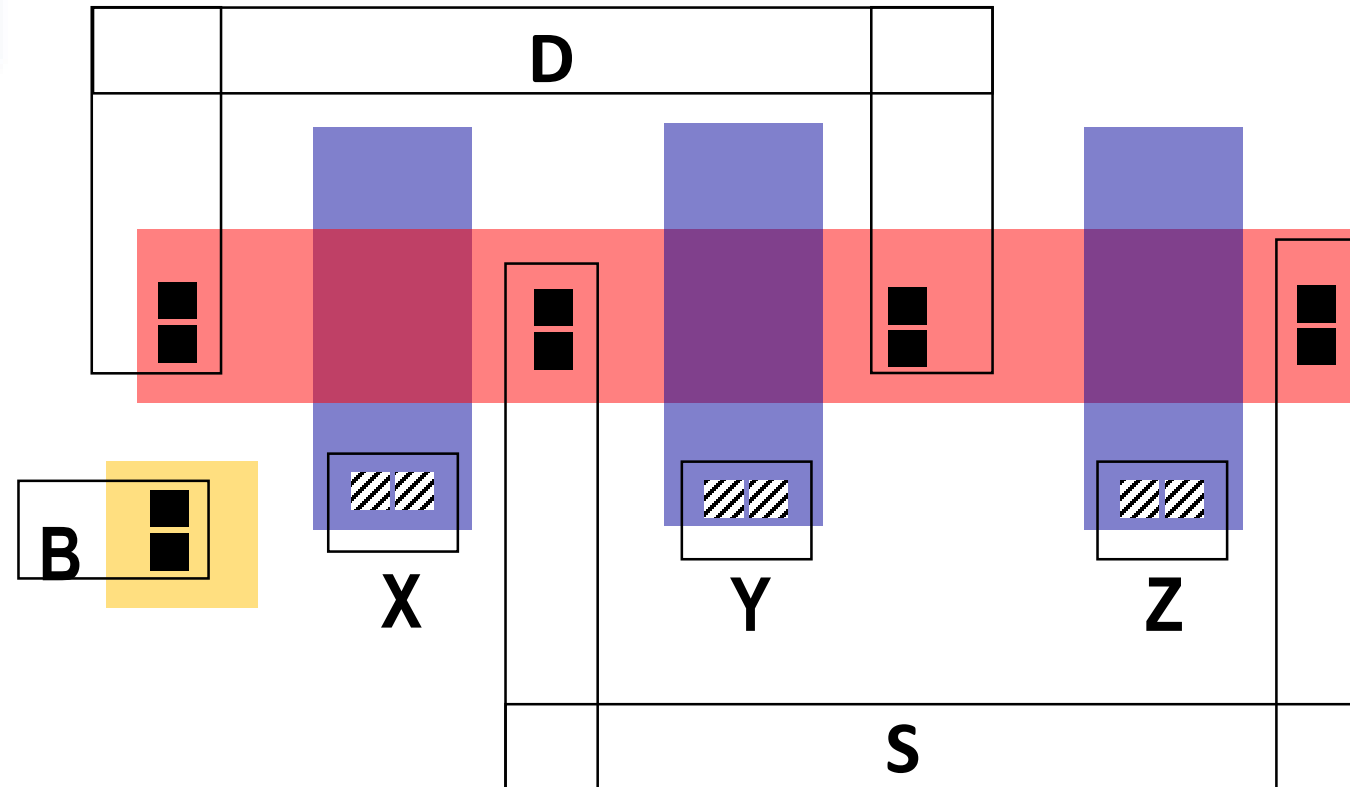
$$(\frac{W}{L})_X = (\frac{W}{L})_Y$$



Parallel MOSFETs (2/2)



$$\left(\frac{W}{L}\right)_X = \left(\frac{W}{L}\right)_Y = \left(\frac{W}{L}\right)_Z$$

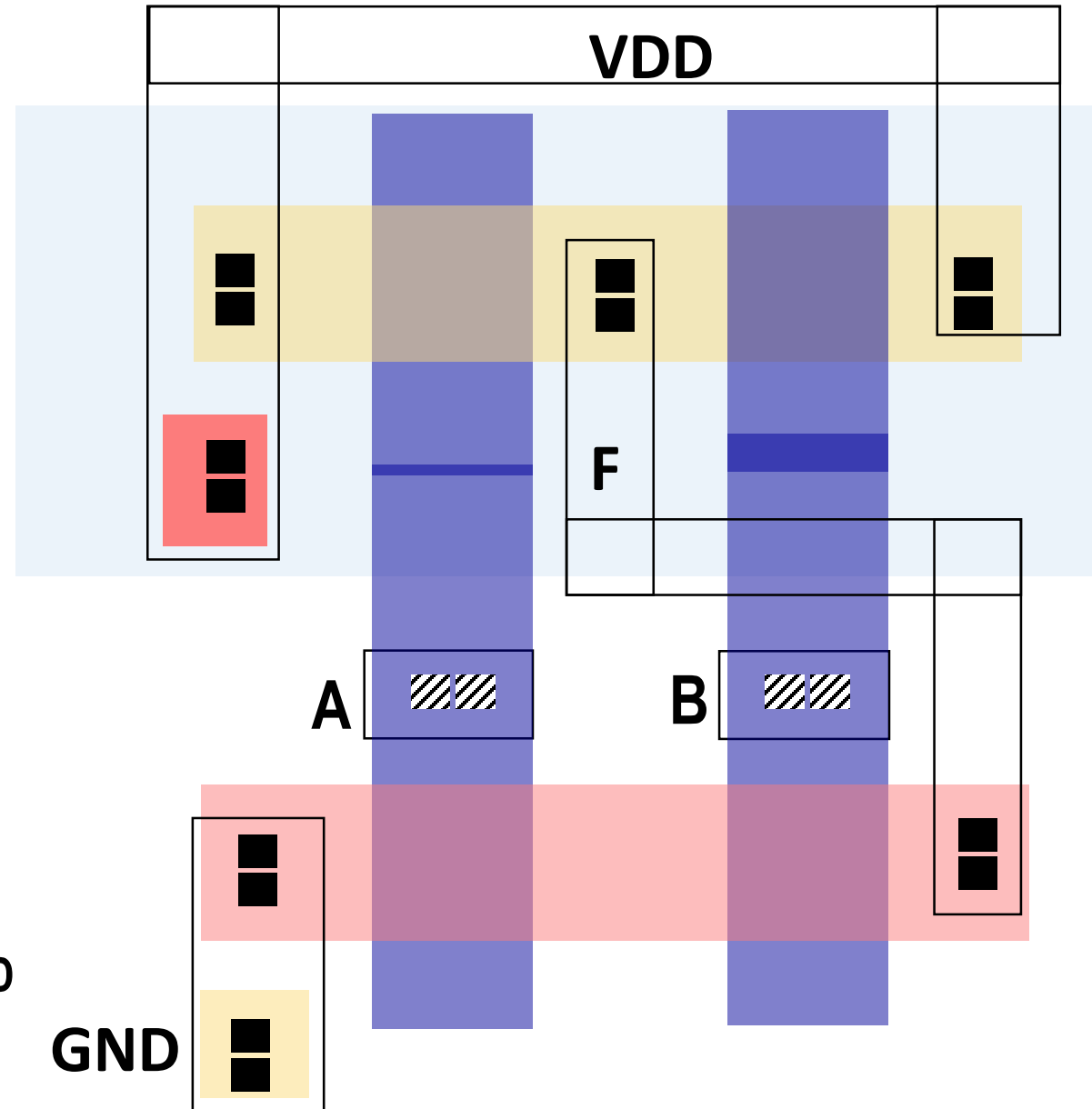
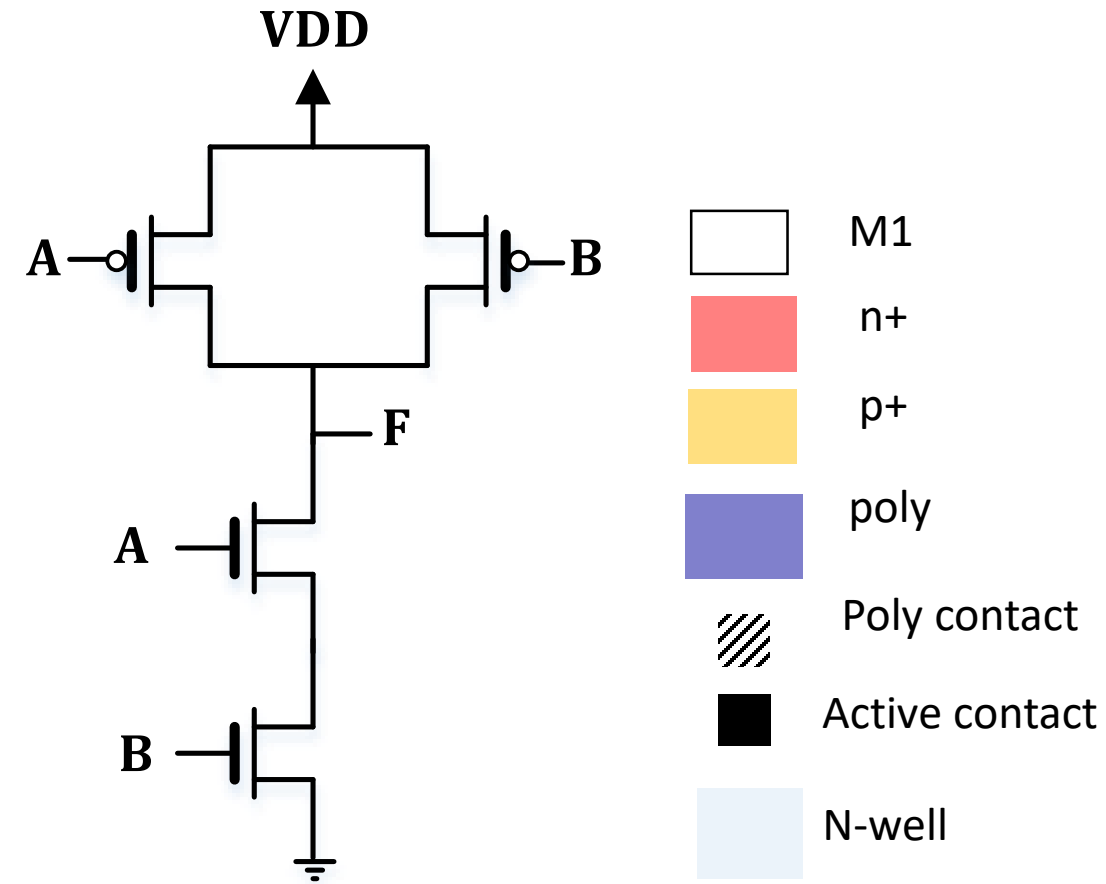


- M1
- n+
- p+
- poly
- Poly contact
- Active contact



- ☐ Layout Methodologies
- ☐ Circuit Elements
- ☐ Resistor
- ☐ Capacitor
- ☐ Inductor
- ☐ MOSFET
- ☐ CMOS Inverter
- ☐ Series MOSFETs
- ☐ Parallel MOSFETs
- ☐ **NAND Gate**
- ☐ NOR Gate

NAND Gate



Note:

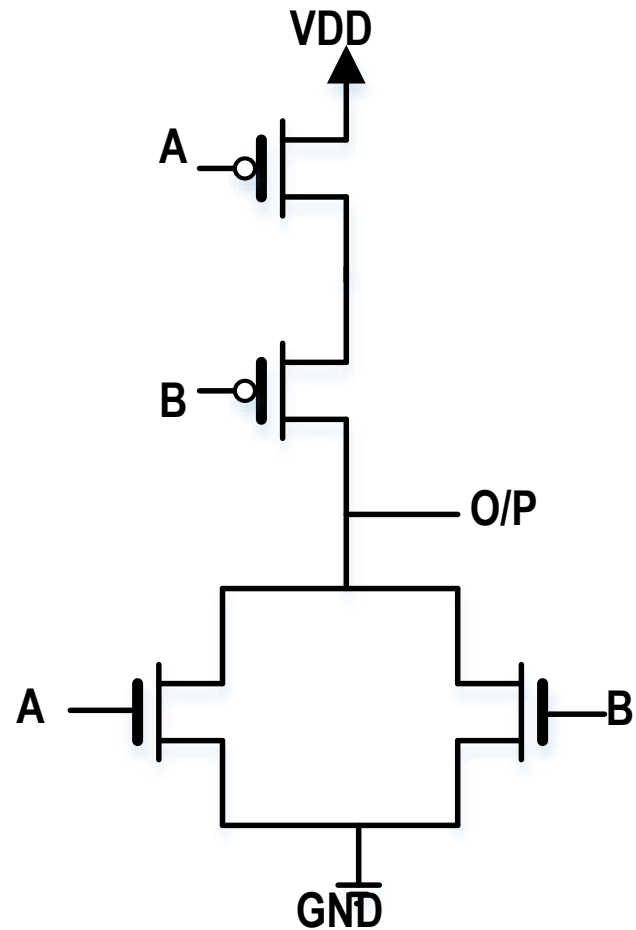
Bulk/Substrate of P-MOSFETs are connected to VDD

Bulk/Substrate of n-MOSFET are connected to GND



- ☐ Layout Methodologies
- ☐ Circuit Elements
- ☐ Resistor
- ☐ Capacitor
- ☐ Inductor
- ☐ MOSFET
- ☐ CMOS Inverter
- ☐ Series MOSFETs
- ☐ Parallel MOSFETs
- ☐ NAND Gate
- ☐ **NOR Gate**

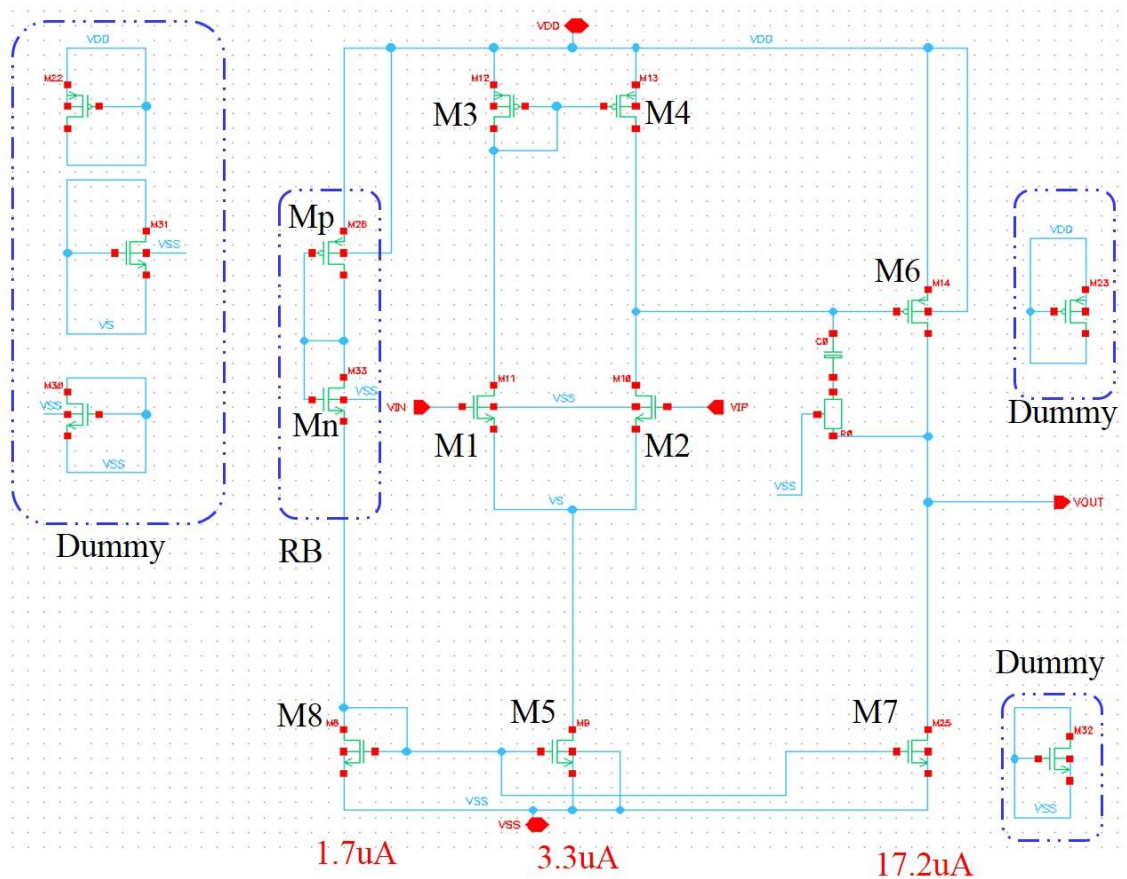
NOR Gate



Note:

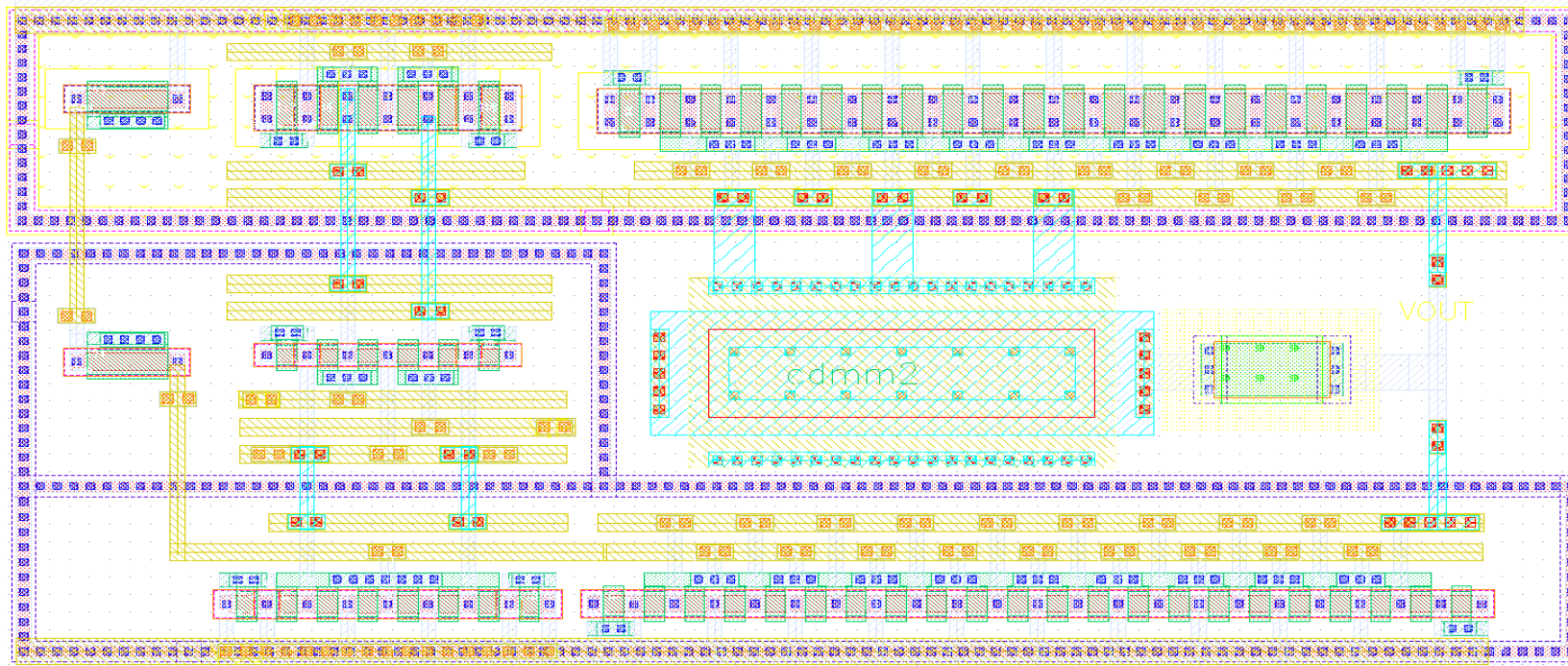
Bulk/Substrate of P-MOSFETs are connected to VDD

Bulk/Substrate of n-MOSFET are connected to GND



2009/2010

Main specifications	Required specification	Schematic Simulation results	Post-layout Simulation results
DC-gain	> 60	99.35 dB	97.59 dB
GBW	>1 MHz	13.5 MHz	12.57 MHz
SR	> 10 $v/\mu s$	13 $v/\mu s$	12.62 $v/\mu s$
Phase Margin	>60°	65°	63.72 °
CMRR	-	107.8 dB	107.7 dB
PSRR+	-	103.7 dB	94.57 dB
PSRR-	-	138.5 dB	100.3 dB
Total current	-	22.3 μA	22.3 μA
Total Power consumption	-	73.5 μW	73.5 μW
FOM=GBW _(Mhz) *CL _(pf) /I _{Tot} (mA)		605.38	560.5
FOM=SR _(V/μs) *CL _(pf) /I _{Tot} (mA)		582.95	565
Area	80 μm * 40 μm		



```
#####
##          CALIBRE SYSTEM          ##
##          LVS REPORT              ##
#####

REPORT FILE NAME:  OTA_CIRCUIT_2STAGE.lvs.report
LAYOUT NAME:      /home/ilotfy/IBMO.13/OTA_CIRCUIT_2STAGE.sp ('OTA_CIRCUIT_2STAGE')
SOURCE NAME:      /home/ilotfy/IBMO.13/OTA_CIRCUIT_2STAGE.netlist.lvs ('OTA_CIRCUIT_2STAGE')
RULE FILE:        /home/ilotfy/IBMO.13/cmr?8sf.lvs.cal_
CREATION TIME:    Sun Apr 15 02:58:03 2012
CURRENT DIRECTORY: /home/ilotfy/IBMO.13
USER NAME:        ilotfy
CALIBRE VERSION:  v2009_2.18.12   Thu May 14 15:08:00 PDT 2009

OVERALL COMPARISON RESULTS

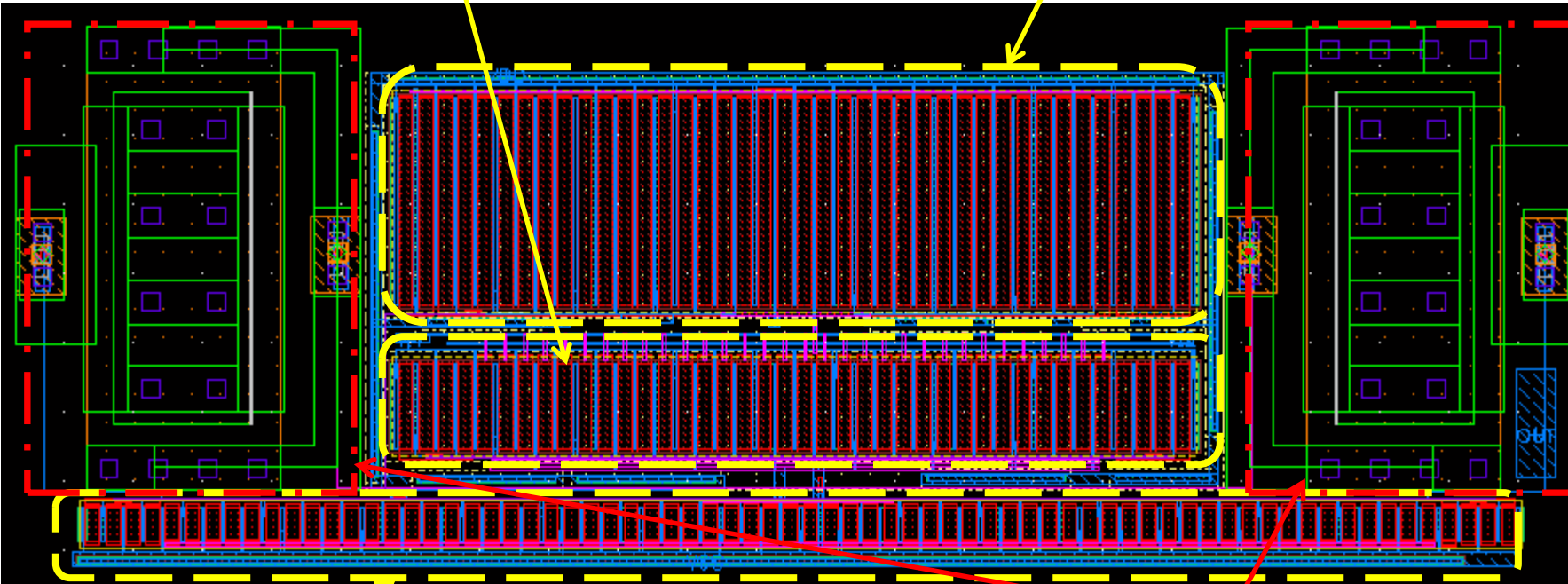
#####
#          CORRECT          #
#####
```

Layout Cell / Type	Source Cell	Nets	Instances	Ports
subc	subc	0L, 0S	0L, 0S	0L, 0S
OTA_CIRCUIT_2STAGE	OTA_CIRCUIT_2STAGE	13L, 13S	13L, 13S	6L, 6S

- ❑ This layout consumes a small area equals to $80\text{ }\mu\text{m} \times 40\text{ }\mu\text{m}$.
- ❑ This work is passed DRC and LVS as shown in following image.

Differential pair

Current Mirror with Dummies



Active Load

Capacitor Cc

```
#####
##          CALIBRE SYSTEM          ##
##          LVS REPORT              ##
#####

REPORT FILE NAME:  OTA_CIRCUIT_2STAGE.lvs.report
LAYOUT NAME:      /home/ilotfy/IBM0.13/OTA_CIRCUIT_2STAGE.sp ('OTA_CIRCUIT_2STAGE')
SOURCE NAME:      /home/ilotfy/IBM0.13/OTA_CIRCUIT_2STAGE.netlist.lvs ('OTA_CIRCUIT_2STAGE')
RULE FILE:        /home/ilotfy/IBM0.13/cmr?8sf.lvs.cal_
CREATION TIME:    Sun Apr 15 02:58:03 2012
CURRENT DIRECTORY: /home/ilotfy/IBM0.13
USER NAME:        ilotfy
CALIBRE VERSION:  v2009_2_18.12   Thu May 14 15:08:00 PDT 2009

OVERALL COMPARISON RESULTS

#####
#          CORRECT          #
#####
```

File	View	Highlight	Tools	Window	Setup	Help
Comparison Results x						
Layout Cell / Type	Source Cell	Nets	Instances	Ports		
subc	subc	0L, 0S	0L, 0S	0L, 0S		
OTA_CIRCUIT_2STAGE	OTA_CIRCUIT_2STAGE	13L, 13S	13L, 13S	6L, 6S		

- ❑ This layout consumes a small area equals to $114 \mu\text{m} \times 40 \mu\text{m}$.
- ❑ This work is passed DRC and LVS as shown in following image.

Thanks!